



UNIVERSITAS
GADJAH MADA

Assignment 1

Single Layer Perceptron for Iris Binary Classification

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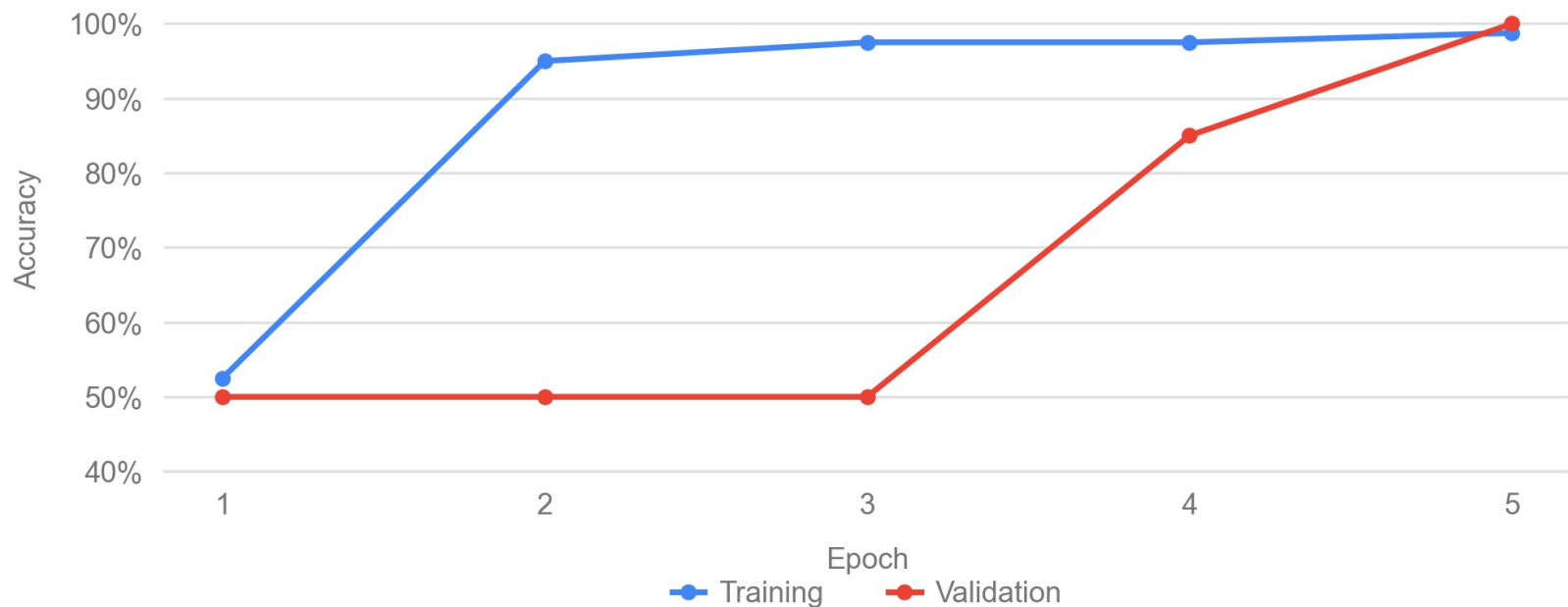
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Deep Learning — Universitas Gadjah Mada

Accuracy — Google Sheets

Per-epoch accuracy from the two worksheets: 80 training samples, and the 20 held-out samples scored on the weights each epoch ends with.

Accuracy per Epoch — Training vs Validation

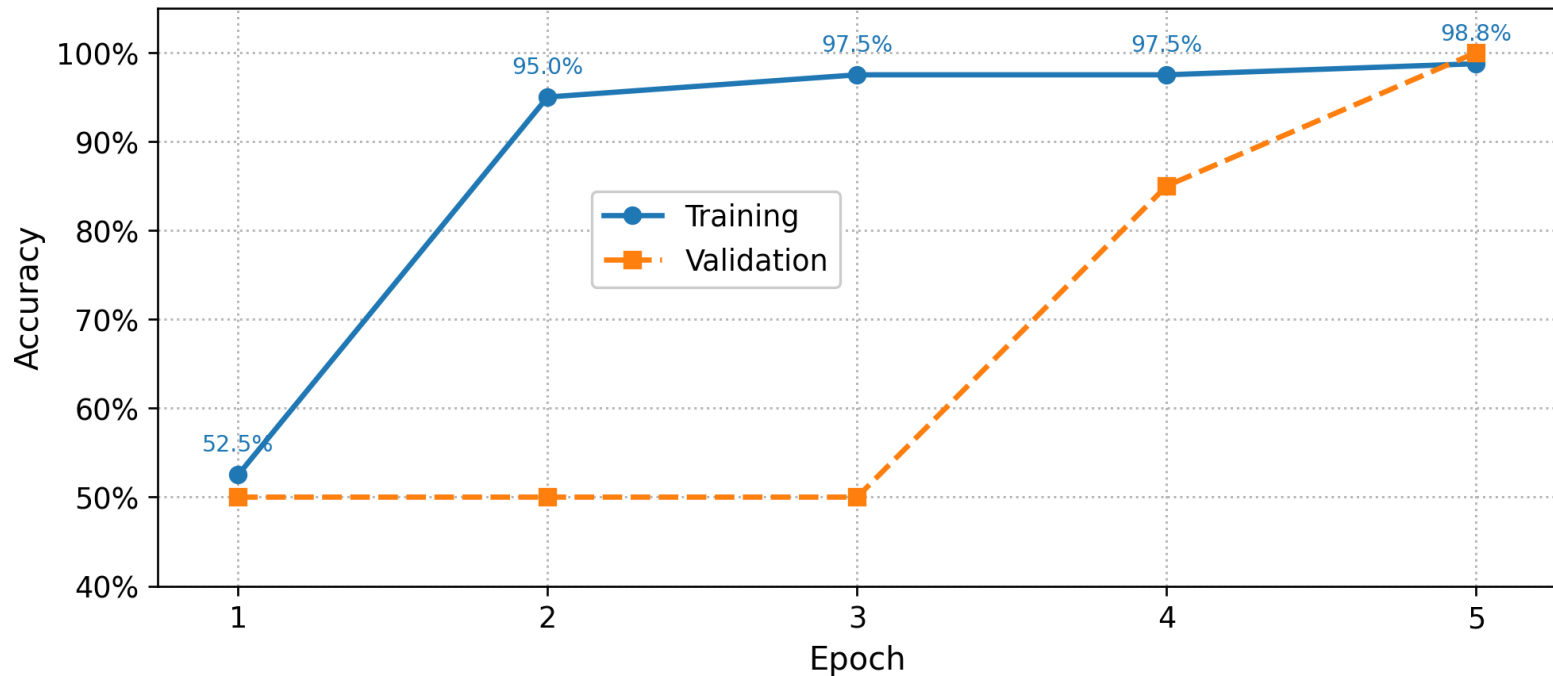


Training accuracy rises 52.50% → 98.75%. Validation sits at 50.00% while the boundary still leans on sepal size, then reaches 85.00% at epoch 4 and 100.00% at epoch 5 as the petal weights take over.

Accuracy — Python Implementation

The same run reproduced by SLP-python.py — identical initialisation, update rule and sample order.

Accuracy per Epoch — Training vs Validation

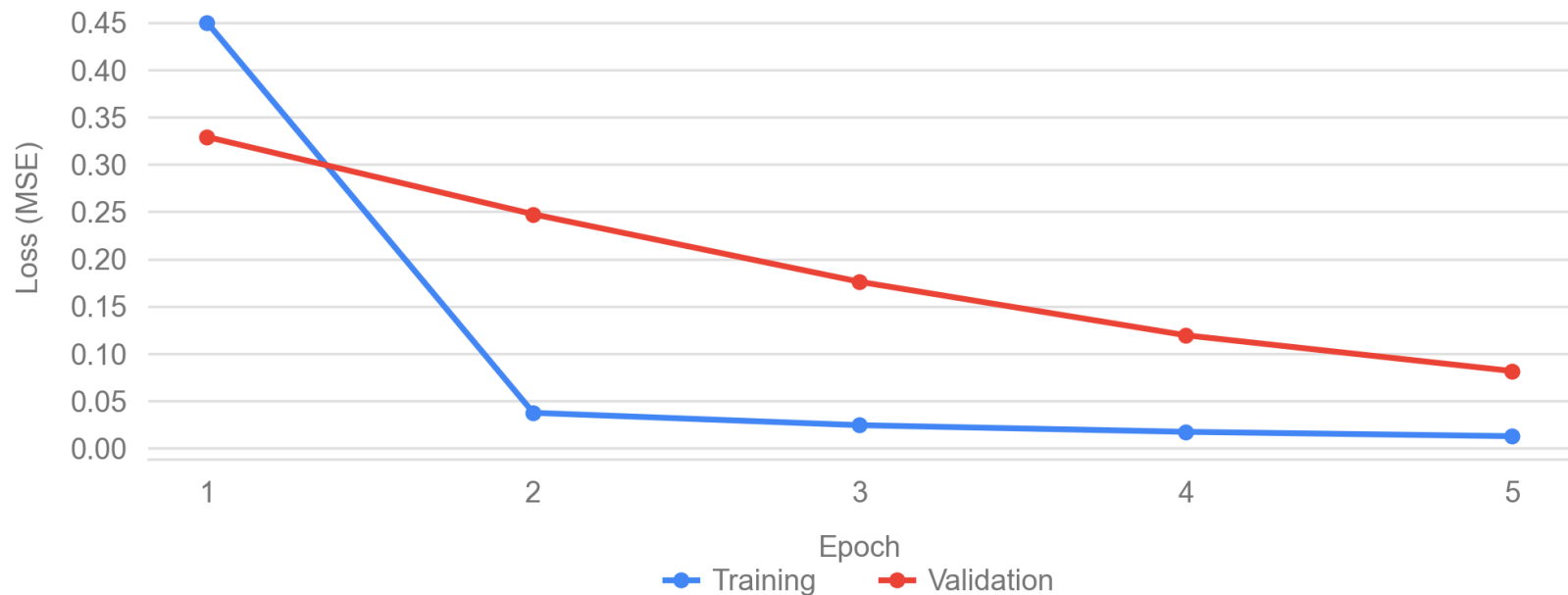


Every epoch matches the spreadsheet exactly: 52.50%, 95.00%, 97.50%, 97.50%, 98.75% (training) and 50.00%, 50.00%, 50.00%, 85.00%, 100.00% (validation).

Loss (MSE) — Google Sheets

Mean Sum-Square-Error per epoch. Validation reuses the bias and teta that training ends each epoch on, exactly as the worksheet specifies.

Loss (MSE) per Epoch — Training vs Validation

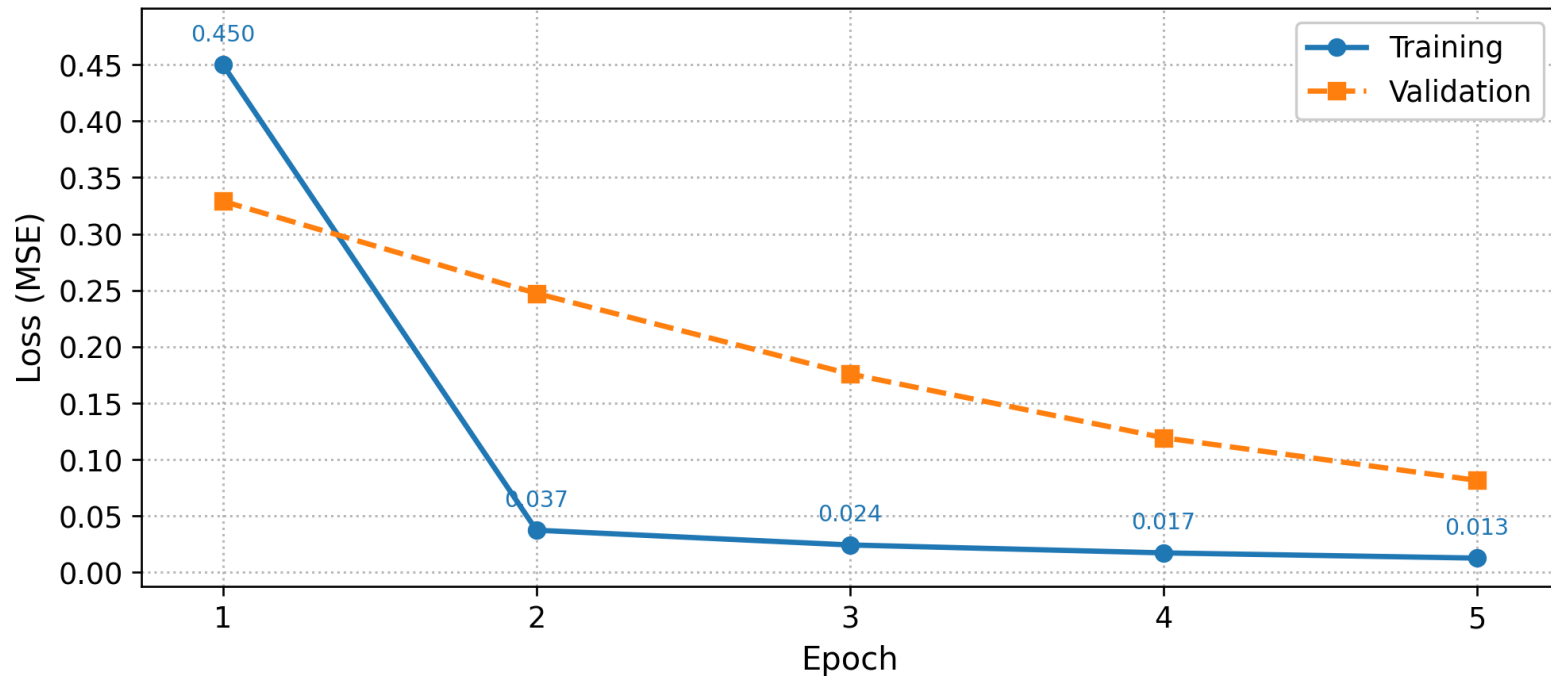


Training loss falls $0.449889 \rightarrow 0.012740$, the steepest drop in epoch 2. Validation loss decreases steadily $0.328951 \rightarrow 0.081581$ — it never turns upward, so there is no overfitting across these five epochs.

Loss (MSE) — Python Implementation

Loss curves produced by SLP-python.py for the same five epochs.

Loss (MSE) per Epoch — Training vs Validation



Training loss reproduces the worksheet's Sum-Square-Error column exactly — 0.4498886551, 0.0374519142, 0.0243721950, 0.0173572017, 0.0127401582 — confirming both implementations follow the same update rule.

Source Links

Spreadsheet and source code for Tasks 1 and 2.

01 Google Spreadsheet

Task 1 — SLP training and validation worksheets, 5 epochs, 80 training / 20 validation samples

<https://docs.google.com/spreadsheets/d/1zPx4ckdLXXAYSfJ87rMhDIB8mPoLJA7x/edit?usp=sharing&oid=118399330202018473068&rtpof=true&sd=true>

02 GitHub Repository

Task 2 — SLP-python.py, the Python implementation of the same model

<https://github.com/muhajav/SLP-Deep-Learning>

References

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6. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
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